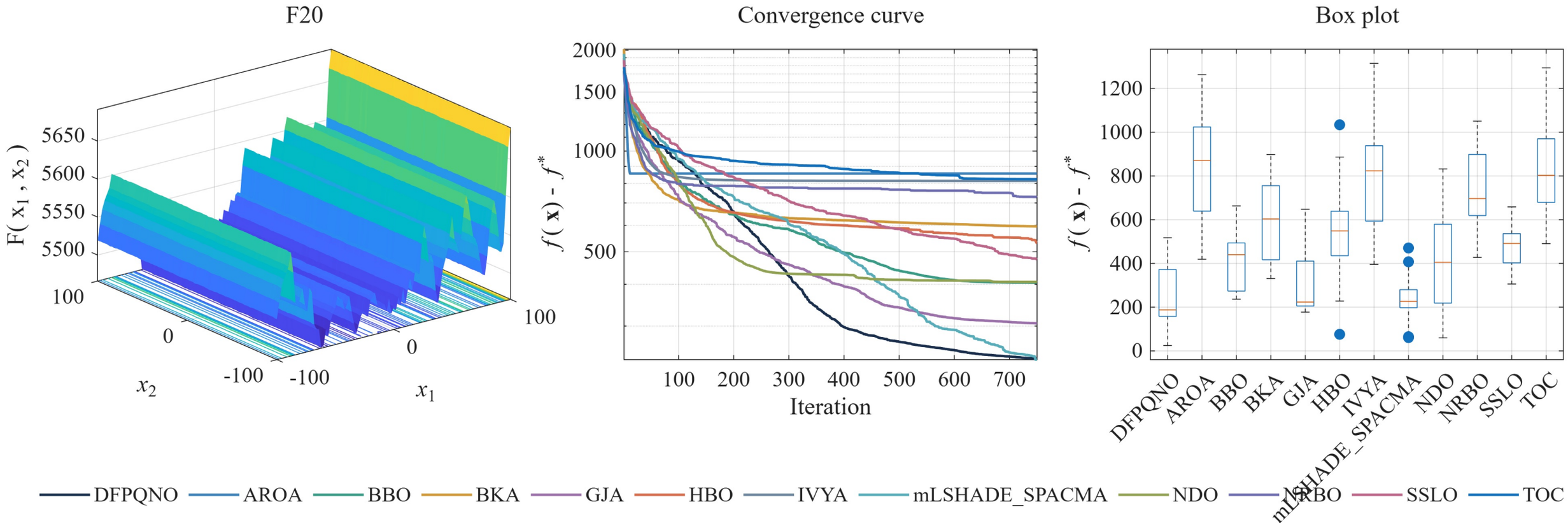

F20
F(x1 , x2)
5650
5600
5550
5500
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
2000
1500
1000
500
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
1200
1000
800
600
400
200
0
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC